

HAITHEM KHACHLOUF

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AI Engineer — Actively Seeking a Full-Time Position (CDI)

Computer Sciences Engineer (ENSI) with hands-on experience building multi-agent LLM systems, memory-augmented AI frameworks, and full-stack AI-driven platforms through internships and academic projects. Curious, adaptable, and motivated to tackle real-world engineering challenges while continuously learning new technologies.

EXPERIENCE

AI Engineering Intern (Final Year Project) — KPIT Engineering

Feb–Jun 2026, Sfax

Designed and implemented a **Memory-Augmented Multi-Agent LLM Framework** for automated automotive test case generation.

- Developed Working, Semantic, Episodic, Procedural, Associative, and Knowledge-Base Memory systems.
- Integrated vector retrieval and memory mechanisms using embeddings and similarity search.
- Enhanced a LangGraph-based agentic framework supporting CLI, REST API, UI, sync, async, and batch execution modes.
- Implemented benchmarking and validation pipelines that raised output quality score from 0.30 (baseline, failing) to 0.80 (+**167%**), turning a failing validation verdict into a pass with only a 4.7% increase in execution time.

Technologies: Python, LangGraph, LangChain, OpenAI, FAISS, Docker

AI Intern — Talan Tunisia (SummerCamp)

Jul–Aug 2025, Tunis

Contributed to the development of **RevMA**, a full-stack AI-based market reverse-engineering platform.

- Designed a **Targeting Agent** and multi-agent workflow using **Python**, **FastAPI**, and **Google Gemini Pro**.
- Assisted in frontend development with **Next.js** and **TypeScript**, and database integration using **PostgreSQL/MongoDB**.

Software Development Intern — TFM (Tunisia Fleet Management)

Jun–Jul 2024, Tunis

Developed an **employee management web application** using **Spring Boot** and **Angular**.

- Designed and integrated **RESTful APIs** to ensure smooth interaction between backend and frontend.

PROJECTS

Talan Challenge — AI Corporate Intelligence Dashboard

2025

Built a multi-agent research system generating structured corporate intelligence reports across **8 analysis dimensions** (financials, operations, tech, ESG, and more).

- Designed the agentic research workflow with **LangGraph** and **LangChain**, integrating multiple search and social-data APIs with automatic query refinement.
- Built a **React** dashboard and **FastAPI** backend for real-time report generation and visualization.

Technologies: Python, LangGraph, LangChain, FastAPI, React, Google Gemini

PlanForge-Agent — AI Business Plan & Pitch Deck Generator

2025

Built a 3-agent pipeline (market research, business planning, pitch deck generation) that turns a business idea into a structured plan and pitch deck.

- Orchestrated asynchronous agent communication using the **a2a-sdk** framework and **Google Gemini**.
- Automated generation of investor-ready, structured 10-slide pitch decks exported as PDF.

Technologies: Python, a2a-sdk, Google Gemini, Asyncio, FPDF

CleanGrid-AI — Multi-Agent Simulation Environment

2025

Designed a multi-agent system with probabilistic agent behaviors to study dirt dynamics and emergent cleaning patterns.

- Modeled agent-environment interactions and analyzed emergent behavior through simulation and visualization.

Technologies: Python, NumPy, Random, Matplotlib

SKILLS

Agentic AI & LLMs: LangGraph, LangChain, OpenAI, Google Gemini, Multi-Agent Systems, Memory-Augmented Agents, RAG, Prompt Engineering, Vector Search

Machine Learning: Scikit-learn, Predictive Modeling, Data Analysis, Feature Engineering

Programming Languages: Python, C++, Java, PHP, JavaScript, SQL

Backend: FastAPI, Django, Spring Boot

Web Development: React.js, Angular, Next.js, HTML, CSS

Databases: PostgreSQL, MongoDB, SQLite, SSMS

Tools: Git, GitHub, IntelliJ IDEA, VS Code, Antigravity IDE, Streamlit, Docker

Data & BI: Power BI, SSIS, Hadoop

Blockchain: Metamask, Ganache, Truffle

EDUCATION

Engineering Cycle in Computer Science, ENSI

2023–2026

Preparatory Cycle in Physics and Chemistry, IPEIS Sfax

2021–2023

Languages: Arabic (Native), French (C1), English (C1) | **Qualities:** Perseverant, adaptable, curious, resilient.